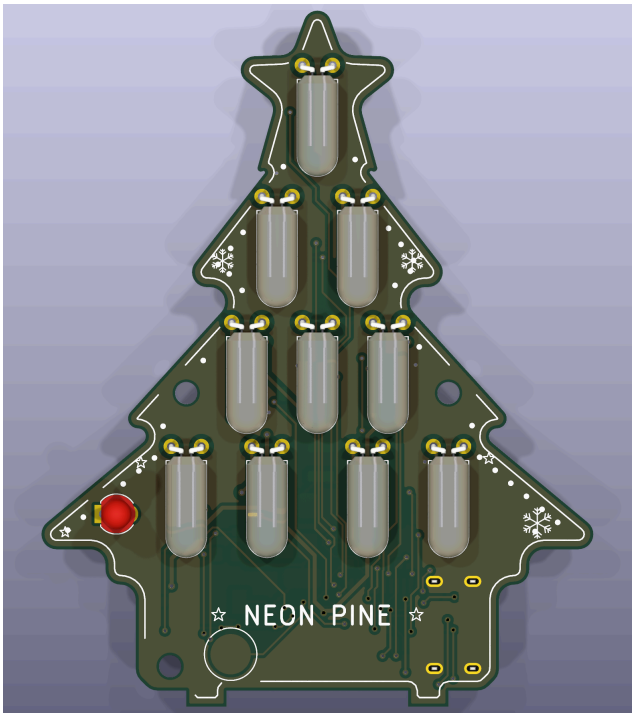
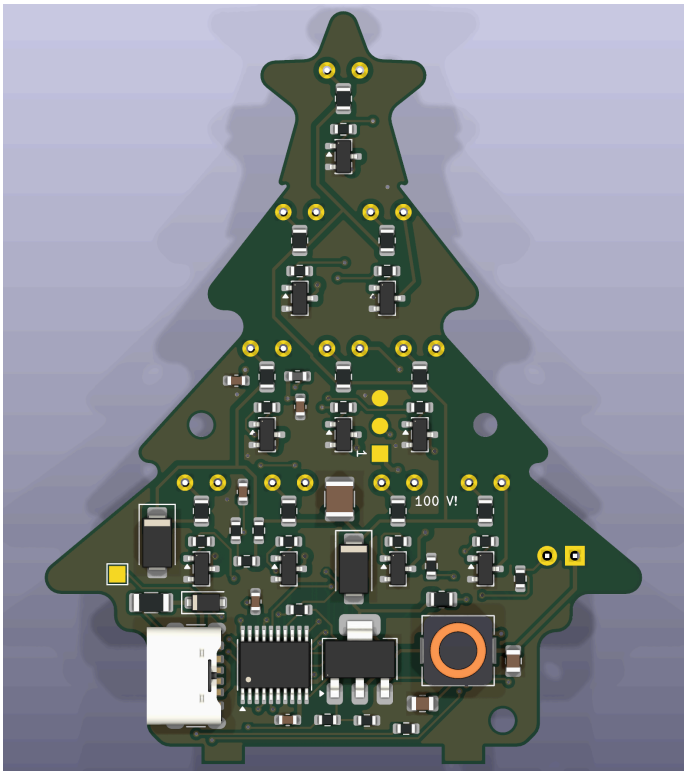
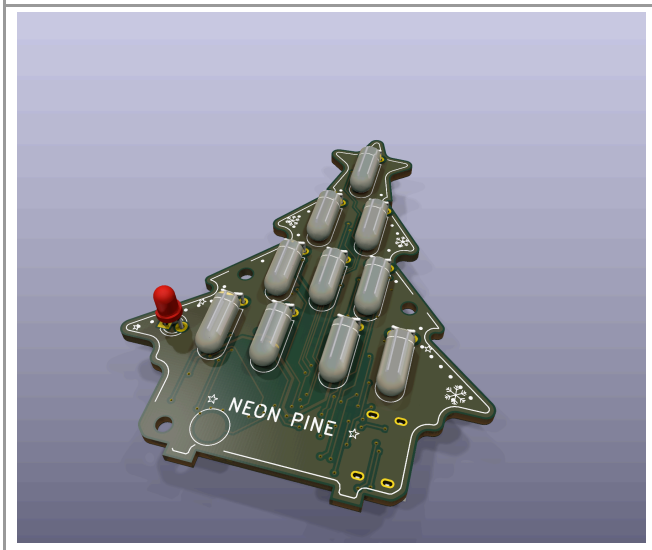
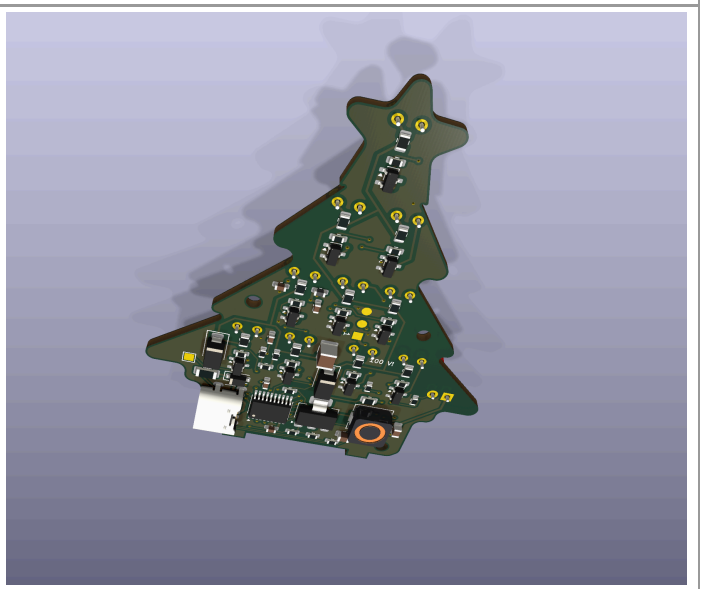
NeonPine – neon-lamp Christmas tree night light

54.8 × 69.1 mm tree-shaped PCB with 10 Brightmaster NEON-3 neon lamps, powered from USB-C 5 V. A CH32V003 MCU runs a 75–105 V boost converter and drives each lamp separately. Rev B adds a screwed-on rear cover (with the product markings) and a slot-in stand base. Unit manufacturing cost ≈ €19 / €9 / €6 at 10 / 100 / 500 pcs (manufacturing §7).

Front	Back (cover removed)
	
	

Rear cover (outer face) and stand base – `NeonPine-cover.kicad_pcb`

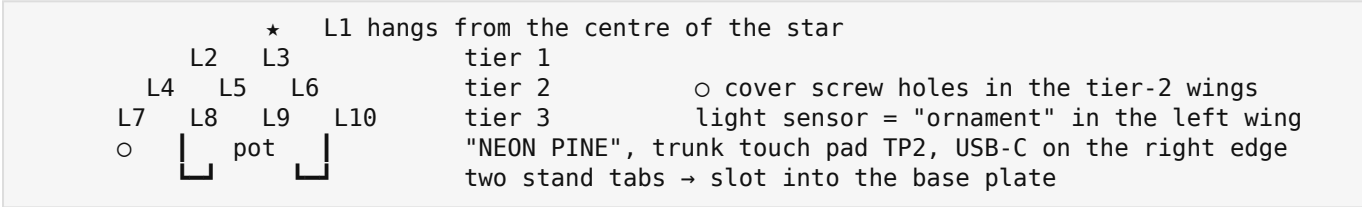


Status (2026-10-06): rev B layout is complete. It is fully routed: 0 DRC violations, 0 unconnected items, and schematic parity shows only cosmetic field differences. All rev A design actions that blocked market release are done in hardware (§7). **Still open before market**

release: firmware, bench and safety verification, the EMC test campaign, supplier declarations, and the company details in the markings (regulatory roadmap).

Document	Contents
docs/NeonPine-schematic.pdf	Schematic (generated)
docs/README.md	Index of all product documentation
docs/compliance/	Regulatory roadmap, risk assessment, EMC test plan, materials, draft DoC / SDoC, labelling, technical file
docs/user-manual.md	User instructions and safety information (draft)
docs/manufacturing.md	PCB spec, assembly incl. cover & stand, programming, end-of-line test, traceability
fab/	Main board: Gerbers + drill, pick & place, BOM, STEP. Cover + stand: Gerbers + drill

1. Concept



- **Lamps:** Brightmaster **NEON-3** (TME). Ø4 × 10 mm, orange, both wire leads out of one end, rated 65 V AC / **90 V DC** maximum striking voltage, 0.5 mA, 20 000 h. Brightmaster does not publish the maintaining voltage, so we assume ≈ 60 V as for an NE-2.
- Footprint `NeonPine:Lamp_Neon_Brightmaster_NEON-3_D4x10_P3.0`: 3.0 mm lead pitch, and the body lies flat on the front, hanging down like an ornament.
- Rev B: the star lamp LA1 hangs from the centre of the star like the others, so its glass stays on the board.
- 3D model from `tools/lamp3d.py`.
- **Sides:** only the lamps, light sensor and trunk touch pad are on the front. Everything else is SMD on the back, under the rear cover.
- **Front art (silkscreen):** a rim line, bead "fairy lights" along the branches, snowflakes, stars and the NEON PINE logotype. It is clipped automatically around pads, holes and parts (`art()` in `tools/pcb.py`).
- **Rear cover:** a plain FR-4 plate in the tree shape, on 3 × M2 × 5 mm standoffs. It covers every HV part and carries the product markings in silkscreen: model, rating, CE, WEEE bin, warnings, manufacturer, S/N field.
- **Stand:** two 4 mm tabs on the bottom edge of the tree and of the cover plug into a 50 × 26 mm base plate. The base comes from the same cover/stand Gerber set.

2. Schematic (PDF)

2.1 Power in

```
USB-C J1 (GCT USB4125-GF-A) VBUS ─┬─ F1 PTC 0.5 A (MF-NSMF050-2) ── VBUS_F ── FB1 600 Ω@100
MHz ──┬─── +5V
CC1 ─R3 5k1─ GND, CC2 ─R4 5k1─ GND D3 SMF5.0A TVS (at the connector)              C1 10µ, C2
100n ┘
```

- +5V feeds U1 (C6 100n at VDD) and the boost (C3 22µ/10V). The ferrite and the caps form a π filter that keeps boost noise off the USB cable.
- The MCU runs straight from 5 V (the CH32V003 is rated 2.7–5.5 V), so the gate drive is a full 5 V.
- The expected maximum input is about 0.1 A.

2.2 HV boost (the MCU is the controller)

```
+5V ── L11 330µH ──SW──┬── D1 ES1D (200 V) ──┬──────────────────┬──────── HV+ (75–105 V)
                       │   (R12 100Ω + C8    C4 1µ/250V D2 SMAJ100A R5 1M − R6 1M ─┬─ VFB →
PD3 (ADC A4)
                 Q1 DMN15H310SE  100p snubber, (1210)   (clamp)                R7 39k ‖ C5
10n
PD2 ─R1 47Ω─ G    (150 V, 330 mΩ @ 5 V)  DNP)  │          │                       │
          R2 100k to GND                      GND        GND                    GND
```

- **Q1 DMN15H310SE:** 150 V, R_DS(on) 330 mΩ at V_GS = 5 V. **R2 is required**: it keeps the boost off during reset and programming. D2 clamps HV+ at V_BR 111–123 V, below Q1's 150 V.
- **R12/C8:** an RC snubber across Q1, not fitted by default (DNP). Fit it if pre-compliance shows SW-node ringing at 30–100 MHz.
- **L11:** 330 µH shielded 6 × 6 × 2.8 mm (SWPA6028S331MT, **verify I_sat ≥ 0.4 A** on the datasheet). Footprint `NeonPine:L_6.3x6.3_H3`, 3D model from `tools/inductor3d.py`.
- **Design point:** 30 kHz in discontinuous conduction mode, P_out max ≈ 0.35 W, I_pk ≈ 0.27 A, t_on ≈ 18 µs at 105 V.
- **Feedback:** 2 M / 39 k gives 105 V → 2.0 V at the ADC. Calibrate against Vrefint. The divider also bleeds C4 after unplugging (τ ≈ 2 s, below 60 V in about 1.1 s).
- **Protection:**
- Firmware: 60 % duty clamp, an overvoltage trip at 110 V, and the IWDG watchdog.
- Hardware: R2 (gate pull-down), D2 (clamp), F1 (PTC), D3 (TVS).
- Firmware limits are **not** counted as safety measures.
- **Stored energy:** at most ½ · 1 µF · 105 V² ≈ 5.5 mJ in C4.

2.3 Lamp channel (× 10)

```
HV+ ── R2n 150k (0805) ──┤NEON LAn├── collector  Qn MMBTA42 (300 V NPN)
                                      base ─ R4n 22k ─ MCU pin (see §2.5)
                                      emitter ─ GND
```

- Lamp current is I = (V_HV − 60 V) / 150 k: 0.31 mA at 105 V, 0.11 mA at 75 V. **The HV rail works as a smooth global dimmer (about 3:1)**, and per-lamp BAM adds detail on top of it.
- **Striking:**
- Firmware starts the rail at ≥ 95 V (a kick to 105 V), and only ramps down once the lamps have struck.
- A lamp that drops out below about 65 V is re-struck with the same kick.

- **Touch current at the lamp leads** is at most 105 V / 150 k = **0.7 mA DC**, because the resistor is on the anode side.
- With DC drive only the cathode glows, so mount the lamp with the cathode electrode facing the viewer.
- **Dark effect:**
- Firmware keeps BAM at 1 kHz or faster with a minimum on-time, plus a 50 ms kick at first ignition.
- Lamps with a radioactive additive (Kr-85) must **not** be used.

2.4 Inputs

Input	HW	Use
TP2 trunk touch pad (front, Ø4 mm, **covered by solder mask**)	R11 1 k to PA2 (TKEY)	tap: next animation · hold: brightness ramp · double-tap: sleep timer · hold 3 s: off/on
Light sensor Q2 TEPT4400 (Vishay, Cd-free NPN phototransistor, front left wing)	+5V → C, E → PD5, R9 22k to GND	auto on at dusk / off in daylight, room-adaptive brightness, wave-to-switch. 20 lx → 0.3–1.5 V, 100 lx → about 4.4 V
TP3 (back pad)	PA1	spare / test point

Rev B removed the star touch pad TP1 and its resistor R10, and replaced the CdS LDR (GL5528, contains cadmium) with the TEPT4400. **Wave-to-switch:** the light sensor is read in the BAM slot where all lamps are dark and again in a lit slot; the difference is the tree's own light reflected off a hand.

2.5 MCU pin map – CH32V003F4P6 (TSSOP-20)

Pin (port)	Function
10–17 (PC0–PC7)	Lamps **L7, L4, L2, L5, L1, L9, L3, L6**. All eight on port C, so one port write per BAM slot
8 (PD0), 1 (PD4)	Lamps **L8, L10**
19 (PD2)	TIM1_CH1 → boost gate (via R1)
20 (PD3)	A4 – HV feedback
2 (PD5)	A5 – light sensor
6 (PA2)	A0 – TP2 trunk touch
5 (PA1)	TP3 spare / test
3 (PD6)	not connected (was TP1)
18 (PD1)	SWIO, on pogo pads J2 (5V / SWIO / GND), back side
4 (PD7)	NRST (C7 100n to GND)

The lamp-to-pin order comes from `tools/pinswap.py` (stored in `tools/lamp_pins.json`). **The firmware maps lamp index → port bit through a table.**

3. Firmware architecture (ch32v003fun or WCH SDK) – *not written yet*

- **Clock:** 48 MHz from HSI × 2. TIM1 runs 30 kHz PWM with **±5 % frequency dithering** (EMC measure E2).
- **Boost loop:** a 1 kHz PI loop driven by the ADC, ramping at most 2 V/ms; the setpoint is the global brightness, 75–105 V. When everything is dark: PWM off and MCU standby.
- **Lamp PWM:** 8-bit bit-angle modulation from TIM2 with an ≈ 1 kHz frame and gamma correction.
- **Animation engine:** each mode is a `frame(t) → uint8_t lvl[10]` function, and modes crossfade over 1.5 s.
- **Single-pad UI on TP2:** tap / hold / double-tap / long hold.
- **Compliance hooks:** an EMC worst-case mode (all lamps on, 105 V), a production self-test mode, and the firmware version readable over SWD.

#	Mode	Idea
1	Candle	independent 1/f flicker per lamp
2	Breathe	all lamps 100 %, only the HV rail moves
3	Twinkle	random lamps swell and decay
4	Garland	spiral chase up the tiers, ending in a star flash
5	Snowfall	soft pulses drifting top → bottom
6	Heartbeat	steady tree, the star double-pulses
7	Drift	auto-cycles 1–6 every 3 min
8	Sleep	dims over 20 min, then off

4. BOM (rev B, approx. LCSC prices)

Full list: fab/NeonPine-bom.csv. R12 and C8 are DNP.

Ref	Part	Qty	≈ $
U1	CH32V003F4P6	1	0.13
LA1–LA10	Brightmaster NEON-3 (TME)	10	*TME price*
Q11–Q20	MMBTA42	10	0.10
Q1	DMN15H310SE	1	0.15
Q2	Vishay TEPT4400 (light sensor)	1	0.10
L11	SWPA6028S331MT 330 µH	1	0.06
D1 / D2 / D3	ES1D / SMAJ100A / SMF5.0A	1+1+1	0.07
C4	1 µF 250 V X7R 1210	1	0.06
R21–R30 / R41–R50	150k 0805 / 22k 0603	10+10	0.03
other R	0603 (47, 100k, 5k1 ×2, 1M ×2, 39k, 22k, 1k)	9	0.02
C1–C7	10µ, 100n ×3, 22µ/10V 0805, 10n, 1µ/250V	7	0.04
FB1	600 Ω @ 100 MHz 0603	1	0.01
J1 / F1	USB4125-GF-A / MF-NSMF050-2	1+1	0.15
mech	rear cover + stand PCB, 3 × M2 × 5 standoff, 6 × M2 × 4 screw	–	*PCB price* + 0.10

Ref	Part	Qty	≈ $
	Total excl. lamps and PCBs (live LCSC prices 2026-10-06)		**$3.17 / $2.47 / $2.15** at 10 / 100 / 500 units

5. PCB & layout

- **Board:** 2 layers, FR-4 1.6 mm, 54.8 × 69.1 mm incl. stand tabs. GND pour on both layers (SMD pads solid, THT pads with thermal spokes, USB shell solid); all vias tented.
- **Netclasses** (`tools/pcb.py` writes them to `NeonPine.kicad_pro`):

Class	Nets	Track	Clearance	Via	Layers
HVBUS	HV+, SW, SNUB (unlimited 105 V source)	0.3 mm	**0.5 mm**	–	**B.Cu only** (enforced in `route.py`, verified: 0 items on the front)
HV	lamp anode / cathode nets, FB_MID (current-limited through ≥ 150 k)	0.3 mm	0.5 mm	0.7/0.35	both
PWR	VBUS, +5V, GND	0.4 mm	0.15 mm	0.7/0.35	both
Default	everything else	0.2 mm	0.15 mm	0.6/0.3	both

- **Placement:**
- Fixed by hand: the lamps and their R/R/Q stacks, J1, U1, Q1, L11, the input protection D3/F1 (right above USB-C), the 3 cover holes and the touch pad.
- The other ≈ 25 parts are placed by the annealing optimiser in `pcb.py`, with weights on the boost loops, decoupling, the ADC filter and the USB input.
- Results: D1 3.2 mm from Q1's drain tab, D1–C4 3.2 mm, C6 at U1 VDD 2.2 mm, the TVS 5.7 mm from VBUS.
- **Routing:** Freerouting **2.4.1** (`tools/route.py`; version 2.1.0 writes incomplete sessions). Freerouting leaves the last 2–3 tight connections open; `tools/fixup.py` closes them with short legal links checked by KiCad's own collision test.
- **Design flow:** 1. `tools/gen.py` writes the schematic. 2. `kicad-cli sch export netlist -o np.net NeonPine.kicad_sch` produces the netlist. 3. `tools/pcb.py np.net` builds the outline, art, placement and pours. 4. `tools/pinswap.py` solves the lamp-to-pin map; if it changed, repeat steps 1–3. 5. `tools/route.py <freerouting-2.4.1.jar>` routes the board. 6. `tools/fixup.py` closes the remaining links; then run DRC. 7. `tools/cover.py` writes the cover and stand board. 8. 3D models come from `tools/lamp3d.py` and `tools/inductor3d.py` (FreeCAD).

6. Ratings (to be confirmed by measurement)

Parameter	Value
Input	5 V ⎓ via USB-C, ≤ 0.15 A (expected ≈ 0.1 A max, standby < 1 mA)
Internal voltage	75–105 V DC, ≤ 123 V under fault (D2 clamp), only on the back under the cover
Environment	indoor, dry, 0–35 °C, IP20

Parameter	Value
Light output	≪ 60 lm (outside the scope of Ecodesign Reg. (EU) 2019/2020)

7. Market-release items

#	Item	Status
1	Rear insulating cover over all HV parts, tool-removable (3 × M2 screws)	**done in rev B** – verify with the IEC test probe and a drop test
2	HV supply (HV+/SW) on the back layer only	**done** – 0 HVBUS copper on the front, no HVBUS vias
3	Lamp supplier declarations: RoHS, REACH, **no Kr-85**	open (purchasing)
3b	Replace the CdS LDR with a Cd-free sensor	**done** – Vishay TEPT4400, R9 → 22k
4	Firmware with an EMC worst-case mode and a production test mode; EMC tests	open
5	Product markings	**done on the cover silkscreen** – fill in company, address and country; the S/N field takes a printed label
6	Lamp retention (silicone dab)	process step in `docs/manufacturing.md`
7	ESD: masked touch pad, TVS on VBUS	**done** (TP2 masked, D3 SMF5.0A); star touch pad removed
8	Part numbers for L11, F1, J1	**chosen** (SWPA6028S331MT, MF-NSMF050-2, USB4125-GF-A) – confirm L11 I_sat
9	Stand	**done** – tabs + base plate (cover/stand board)
10	Bench verification: D2 clamp vs Q1, C4 discharge, input current, temperature rise, touch current	open (risk assessment §5)
12	Price check findings: L11 → SWPA6045S331MT (the 6028 is rated only 300 mA and out of stock), C4 → 1812 footprint (no 1 µF/250 V in 1210), Q1 low stock → CJT04N15 alternative	open – see manufacturing §7.3
11	EMC extras: ferrite (done), snubber footprint (done, DNP), boost dithering (firmware), edge stitching vias (not done; add if pre-scan needs it)	partly